

AgilityCare Health Operations Platform (AHOP)

Source Code & Documentation

Version: 1.0

Date: June 2026

Repository: <https://github.com/jhewhome/ahop-snipe-it>

Live deployment: <http://35.247.142.84>

Local development: <http://localhost/snipe-it/public>

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1. Executive Summary

The **AgilityCare Health Operations Platform (AHOP)** is a healthcare information system built as a customization of the open-source **Snipe-IT** asset management platform (Laravel 11). AHOP extends Snipe-IT with integrated clinical workflows:

- Patient registration and EMR-lite records
- Outpatient Department (OPD) visits and queue
- Appointment scheduling and reception check-in
- Laboratory orders and results
- Billing and payment tracking
- Medical equipment and supplies (via Snipe-IT assets module)
- Clinical dashboard, analytics, and operational reports

The system is deployed on **Google Cloud Platform** for demonstration and supports local development on **XAMPP (Windows)**. Source code is version-controlled on **GitHub**.

2. System Overview

2.1 Purpose

AHOP centralizes clinic operations: patient data, clinical encounters, laboratory workflows, billing, and biomedical equipment tracking in a single web application with role-based access for reception, clinical staff, laboratory, biomedical engineering, and clinic administration.

2.2 Users and Roles

Role group	Primary functions
AHOP Reception	Patients, appointments, check-in, billing (create/view)
AHOP Clinic Staff	OPD documentation, patient updates, appointments
AHOP Laboratory	Lab orders, result entry
AHOP Biomedical	Medical equipment assets and maintenance
AHOP Clinic Administrator	Full clinical and billing access, reports
Superuser (IT)	System settings, user management

2.3 Demo Accounts (after seeding)

Username	Password	Role
clinicadmin	demo1234	Clinic Administrator
reception	demo1234	Reception
physician	demo1234	Clinic Staff
dr.santos	demo1234	Clinic Staff (attending physician)
labtech	demo1234	Laboratory
biomedical	demo1234	Biomedical

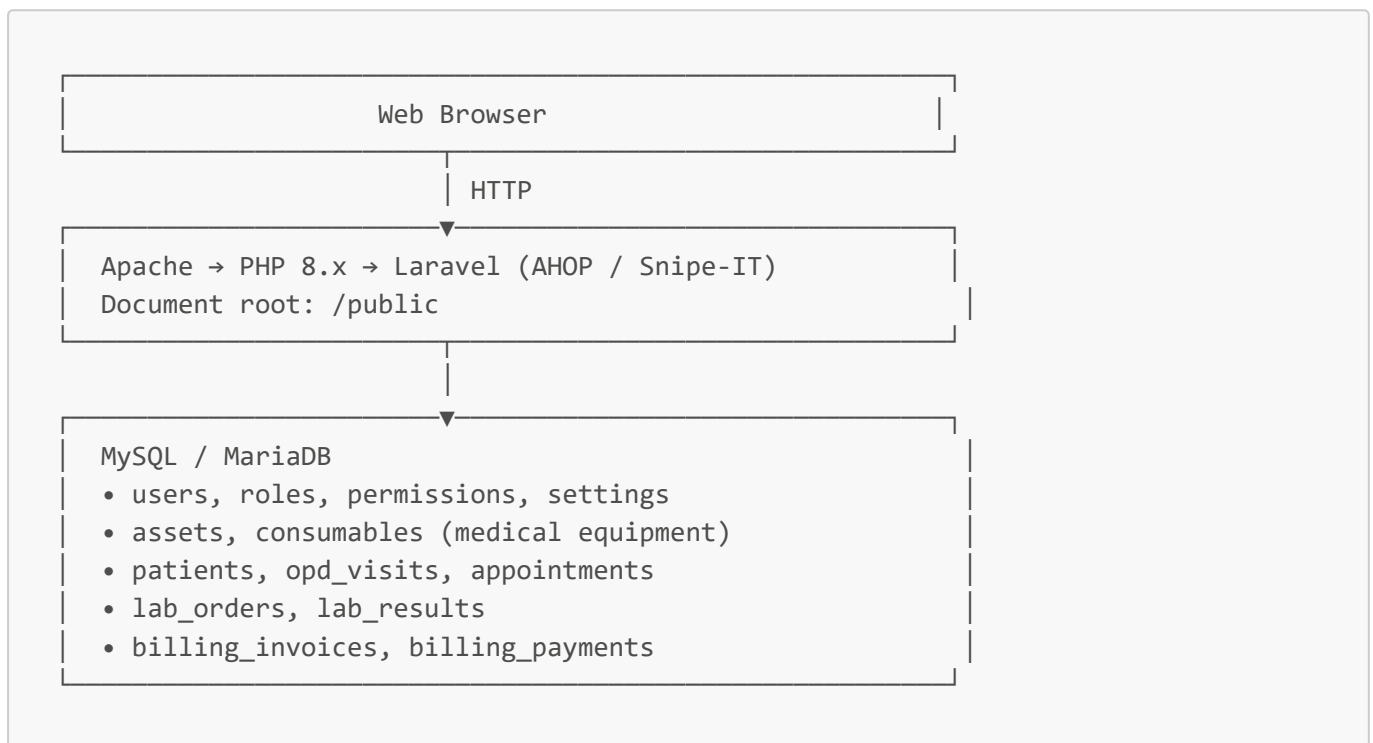
3. Technology Stack

Layer	Technology
Language	PHP 8.2+
Framework	Laravel 11
Base application	Snipe-IT (AGPL-3.0)
Database (default)	MySQL / MariaDB
Database (optional clinical)	PostgreSQL (dual-database mode)
Web server	Apache 2.4 (with <code>mod_rewrite</code>)

Layer	Technology
Frontend	Blade templates, Bootstrap, jQuery, Chart.js
Theme	Custom AHOP CSS (public/css/ahop-theme.css)
Cloud hosting	Google Cloud Platform (Compute Engine VM)
Version control	Git / GitHub

4. System Architecture

4.1 Single-database mode (current deployment)



4.2 Dual-database mode (optional)

When `AHOP_CLINICAL_DATABASE_ENABLED=true`:

- **MySQL** — Snipe-IT core (users, assets, RBAC)
- **PostgreSQL** — Clinical tables (patients, OPD, lab, billing)

Bridge: `app/Support/ClinicalDatabase.php`, trait `UsesClinicalDatabase` on clinical models.

4.3 Request flow (example: OPD visit)

Reception check-in → Patient record → OPD visit created →
Physician documents encounter → Lab order (optional) →
OPD completed → Auto-billing (if enabled) → Invoice & payment

5. Source Code Structure

5.1 Repository layout

```

ahop-snipe-it/
├── app/
│   ├── Console/Commands/           # AHOP artisan commands (seed, backup, setup)
│   ├── Http/Controllers/          # Clinical controllers
│   │   ├── PatientsController.php
│   │   ├── OpdVisitsController.php
│   │   ├── AppointmentsController.php
│   │   ├── LabOrdersController.php
│   │   ├── BillingInvoicesController.php
│   │   ├── ReceptionCheckInController.php
│   │   ├── ClinicalAnalyticsController.php
│   │   ├── ClinicalReportsController.php
│   │   └── Api/LabOrdersController.php
│   ├── Models/                    # Clinical Eloquent models
│   │   ├── Patient.php
│   │   ├── OpdVisit.php
│   │   ├── Appointment.php
│   │   ├── LabOrder.php
│   │   ├── LabResult.php
│   │   ├── BillingInvoice.php
│   │   └── ClinicalModel.php
│   ├── Policies/                  # Authorization policies
│   ├── Services/                  # Business logic
│   │   ├── OpdVisitInvoiceService.php
│   │   ├── AppointmentReminderService.php
│   │   ├── PhysicianSelectService.php
│   │   └── Ahop/ClinicalDashboardService.php
│   └── Support/
│       └── ClinicalDatabase.php    # Dual-DB connection resolver
├── config/
│   ├── ahop.php                   # AHOP feature configuration
│   └── ahop_roles.php              # Role permission templates
├── database/
│   ├── migrations/                 # MySQL migrations (incl. AHOP columns)
│   ├── migrations/clinical/        # Clinical schema migrations
│   └── seeders/                    # Demo data seeders
├── docs/                           # This documentation package
├── public/
│   └── css/ahop-theme.css          # AgilityCare UI theme
├── resources/
│   ├── views/                     # Blade templates (clinical UI)
│   └── lang/en-US/                 # Language strings (ahop.php)
├── routes/
│   ├── web.php                     # Clinical web routes
│   └── api.php                     # REST API (incl. lab integration)
└── .env.example                    # Environment template

```

5.2 Key design patterns

Pattern	Usage
MVC	Laravel controllers, models, Blade views
Service classes	Billing automation, reminders, dashboard metrics
Policies	Per-module authorization (<code>PatientPolicy</code> , <code>OpdVisitPolicy</code> , etc.)
Traits	<code>UsesClinicalDatabase</code> for connection routing
Seeders	Idempotent demo data (<code>ahop:seed-all</code>)
Scheduled tasks	Backups, appointment reminders, equipment alerts

6. Clinical Modules

6.1 Patients (TR5 — centralized patient data)

- **Controller:** `PatientsController`
- **Model:** `Patient`
- **Routes:** `/patients`, `/patients/{id}/clinical-summary`
- **Features:** Registration, demographics, allergies, problem list, company/clinic site

6.2 Appointments (TR2)

- **Controller:** `AppointmentsController`
- **Model:** `Appointment`
- **Routes:** `/appointments`, `/appointments/calendar`
- **Features:** Schedule, calendar view, check-in, email reminders (`ahop:send-appointment-reminders`)

6.3 Reception check-in

- **Controller:** `ReceptionCheckInController`
- **Route:** `/reception/check-in`
- **Features:** Walk-in registration, appointment check-in, patient search

6.4 OPD / EMR-lite (TR1)

- **Controller:** `OpdVisitsController`
- **Model:** `OpdVisit`
- **Routes:** `/opd-visits`, `/opd-visits/queue`
- **Features:** Visit documentation, vitals, assessment, diagnosis, medical certificate, lab order creation

6.5 Laboratory (TR3)

- **Controller:** `LabOrdersController`, `Api\LabOrdersController`
- **Models:** `LabOrder`, `LabResult`
- **Routes:** `/lab-orders`, `/api/v1/lab/orders`
- **Workflow:** ordered → in_progress → completed

6.6 Billing (TR4)

- **Controller:** `BillingInvoicesController`
- **Models:** `BillingInvoice`, `BillingLineItem`, `BillingPayment`, `BillableService`
- **Routes:** `/billing-invoices`
- **Features:** Invoices, line items, payments, receipts, auto-bill on OPD complete

6.7 Medical equipment (TR6, TR7)

- **Base:** Snipe-IT Assets, Consumables, Maintenances modules
- **Seeder:** `MedicalEquipmentSeeder`
- **Features:** Equipment registry, maintenance alerts, dashboard equipment status

6.8 Dashboard & reports (TR12)

- **Controller:** `DashboardController`, `ClinicalReportsController`, `ClinicalAnalyticsController`
- **Views:** `dashboard-ahop.blade.php`
- **Features:** Real-time stats (60s refresh), clinical CSV reports, analytics charts

7. Database Design

7.1 Clinical tables (MySQL or PostgreSQL)

Table	Description
<code>patients</code>	Patient demographics and clinical summary fields
<code>appointments</code>	Scheduled visits, reminder tracking
<code>opd_visits</code>	Outpatient encounters, vitals, documentation
<code>lab_orders</code>	Laboratory test orders
<code>lab_results</code>	Result values and flags
<code>billable_services</code>	Service fee catalog
<code>billing_invoices</code>	Patient invoices
<code>billing_line_items</code>	Invoice line items
<code>billing_payments</code>	Payment records

7.2 Snipe-IT core tables (MySQL)

Table	Description
<code>users</code>	Staff accounts
<code>permission_groups</code>	AHOP role groups
<code>assets</code>	Medical equipment
<code>consumables</code>	Supplies
<code>companies</code>	Clinic / site (Company field)

Table	Description
<code>settings</code>	Application settings
<code>action_logs</code>	Audit trail

7.3 Migration paths

```
# Core + AHOP MySQL migrations
php artisan migrate --force

# Clinical schema (patients, OPD, lab)
php artisan migrate --path=database/migrations/clinical --force

# Optional PostgreSQL setup
php artisan ahop:clinical-db-setup
```

8. Security & Access Control

8.1 RBAC (TR8)

- Role templates: `config/ahop_roles.php`
- Setup command: `php artisan ahop:setup-priority1`
- Permissions: `config/permissions.php` (*patients*, *opd_visits*, *lab_orders*, *billing_invoices*.)
- Policies enforce access per module

8.2 Authentication

- Session-based login (`/login`)
- Username + password (bcrypt hashed)
- Optional: LDAP, SAML, Google OAuth (Snipe-IT built-in)

8.3 Audit logging (TR9)

- Model change history via Snipe-IT `Loggable` trait → `action_logs` table
- Dedicated BHC-style audit UI not included; logs available in admin action logs

8.4 Transport security

- Production: HTTPS recommended (Let's Encrypt or load balancer)
- Demo deployment: HTTP with `APP_FORCE_TLS=false`

8.5 Backup (TR10)

- Command: `php artisan ahop:backup`
- Schedule: daily via `php artisan schedule:run`
- Health check: `php artisan ahop:backup-health`

9. API & Integration

9.1 Laboratory REST API (TR13 — partial)

Method	Endpoint	Description
GET	<code>/api/v1/lab/orders</code>	List lab orders
GET	<code>/api/v1/lab/orders/{id}</code>	Order detail
POST	<code>/api/v1/lab/orders/{id}/results</code>	Submit results

Authentication: Snipe-IT API tokens (Bearer).

9.2 Dashboard JSON

Method	Endpoint	Description
GET	<code>/ahop/dashboard-data</code>	Clinical dashboard refresh payload

10. Artisan Commands

Command	Purpose
<code>ahop:seed-all</code>	Full demo: roles, users, equipment, clinical data
<code>ahop:seed-demo-users</code>	Demo staff accounts
<code>ahop:seed-physicians</code>	Attending physician roster
<code>ahop:seed-equipment</code>	Medical equipment taxonomy
<code>ahop:seed-demo</code>	Patients, OPD, appointments, lab, billing
<code>ahop:setup-priority1</code>	RBAC groups, backup checklist
<code>ahop:clinical-db-setup</code>	PostgreSQL clinical database
<code>ahop:backup</code>	Full backup (MySQL + uploads + optional PostgreSQL)
<code>ahop:send-appointment-reminders</code>	Email reminders (scheduled)
<code>snipeit:create-admin</code>	Create superuser account

11. Configuration Reference

Key entries in `.env` (see `.env.example` for full list):

```
AHOP_CLINICAL_SIDEBAR=true
AHOP_THEME_ENABLED=true
AHOP_CLINICAL_DASHBOARD=true
AHOP_AUTO_BILL_OPD_COMPLETE=true
AHOP_CLINICAL_DATABASE_ENABLED=false
```



```
AHOP_DAILY_BACKUP=true
AHOP_APPOINTMENT_REMINDERS=true
APP_URL=http://35.247.142.84
```

Feature flags map to [config/ahop.php](#).

12. Deployment Summary

Environment	URL	Database
Local (XAMPP)	http://localhost/snipe-it/public	MySQL snipeit
Google Cloud	http://35.247.142.84	MariaDB snipeit
Source code	https://github.com/jhewhome/ahop-snipe-it	—

Deploy workflow: Local edit → [git push](#) → Cloud [git pull](#) → [composer install](#) → [migrate](#) → [config:cache](#)

See [AHOP-DEPLOYMENT-GUIDE.md](#) for step-by-step instructions.

13. Requirements Traceability

TR	Requirement	AHOP implementation	Status
TR1	EMR	OPD visits, patient EMR fields, medical certificate	Partial (clinic-scoped)
TR2	Appointments	Appointments module, calendar, check-in	Implemented
TR3	Laboratory	Lab orders, results, API	Implemented
TR4	Billing	Invoices, payments, auto-bill	Implemented
TR5	Centralized database	MySQL clinical schema	Implemented
TR6	Inventory	Snipe-IT consumables/assets	Implemented
TR7	Equipment monitoring	Asset maintenance, alerts	Implemented
TR8	RBAC	AHOP role groups, policies	Implemented
TR9	Audit logging	Snipe-IT action_logs	Partial
TR10	Backup / DR	ahop:backup , scheduled	Partial (local backup)
TR11	Appointment confirmation	Email reminders	Partial (email only)
TR12	Dashboard	Clinical dashboard, reports	Implemented
TR13	Integration layer	Lab REST API	Partial

14. Source Code Inventory

14.1 AHOP-specific files (custom development)

Configuration

- `config/ahop.php`
- `config/ahop_roles.php`

Controllers (clinical)

- `app/Http/Controllers/PatientsController.php` (extended)
- `app/Http/Controllers/OpdVisitsController.php`
- `app/Http/Controllers/AppointmentsController.php`
- `app/Http/Controllers/LabOrdersController.php`
- `app/Http/Controllers/BillingInvoicesController.php`
- `app/Http/Controllers/ReceptionCheckInController.php`
- `app/Http/Controllers/ClinicalAnalyticsController.php`
- `app/Http/Controllers/ClinicalReportsController.php`
- `app/Http/Controllers/StaffGuideController.php`
- `app/Http/Controllers/Api/LabOrdersController.php`

Models

- `app/Models/Patient.php`
- `app/Models/OpdVisit.php`
- `app/Models/Appointment.php`
- `app/Models/LabOrder.php`
- `app/Models/LabResult.php`
- `app/Models/BillingInvoice.php`
- `app/Models/BillingLineItem.php`
- `app/Models/BillingPayment.php`
- `app/Models/BillableService.php`
- `app/Models/ClinicalModel.php`

Services

- `app/Services/OpdVisitInvoiceService.php`
- `app/Services/AppointmentInvoiceService.php`
- `app/Services/AppointmentReminderService.php`
- `app/Services/AppointmentCheckInService.php`
- `app/Services/PhysicianSelectService.php`
- `app/Services/ClinicSiteService.php`
- `app/Services/Ahop/ClinicalDashboardService.php`
- `app/Services/Ahop/ClinicalReportService.php`
- `app/Services/Ahop/EquipmentAlertService.php`

Console commands

- `app/Console/Commands/SeedAhopAll.php`

- app/Console/Commands/SeedAhopDemo.php
- app/Console/Commands/SeedAhopDemoUsers.php
- app/Console/Commands/SeedAhopPhysicians.php
- app/Console/Commands/SeedMedicalEquipment.php
- app/Console/Commands/SetupClinicalDatabase.php
- app/Console/Commands/SetupPriority1.php
- app/Console/Commands/AhopBackup.php

Database

- database/migrations/clinical/*.php
- database/migrations/2026_05_*_ahop_*.php
- database/seeder/Ahop*.php
- database/seeder/MedicalEquipmentSeeder.php

Views & assets

- resources/views/dashboard-ahop.blade.php
- resources/views/patients/, opd_visits/, appointments/, lab_orders/, billing_invoices/, reception/
- resources/views/partials/ahop-*.blade.php
- resources/views/staff-guide/
- public/css/ahop-theme.css
- resources/lang/en-US/ahop.php

14.2 Inherited from Snipe-IT (not modified for thesis inventory)

Standard Snipe-IT modules remain available when clinical sidebar mode is off: licenses, accessories, components, kits, depreciation, LDAP, etc.

Document history

Version	Date	Changes
1.0	June 2026	Initial thesis documentation package

End of document — AgilityCare Health Operations Platform (AHOP)